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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 4.5 k-fold cross-validationStochastic for Gradient Descent (SDG) using the MNIST
data set
5  Machine Learning for Engineering Problem Solving
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7  """
8
9  import IPython as IP
10 IP.get_ipython().magic('reset -sf')
11
12 import numpy as np
13 import scipy as sp
14 import pandas as pd
15 from scipy import fftpack, signal # have to add
16 import matplotlib as mpl
17 import matplotlib.pyplot as plt
18 import sklearn as sk
19 from sklearn import linear_model
20 from sklearn import pipeline
21 from sklearn import datasets
22
23 cc = plt.rcParams['axes.prop_cycle'].by_key()['color']
24 plt.close('all')
25
26
27 %% Load your data
28
29 # Fetch the MNIST dataset from openml
30 mnist = sk.datasets.fetch_openml('mnist_784',as_frame=False)
31 X = mnist['data'] # load the data
32 Y = np.asarray(mnist['target'],dtype=int) # load the target
33
34 # Split the data set up into a training and testing data set
35 X_train = X[0:60000,:]
36 X_test = X[60000:,:]
37 Y_train = Y[0:60000]
38 Y_test = Y[60000:]
39
40 %% Train a Stochastic Gradient Descent classifier
41
42 # Extract a subset for our "5-detector".
43 Y_train_5 = (Y_train == 5)
44 Y_test_5 = (Y_test == 5)
45
46 sgd_clf = sk.linear_model.SGDClassifier()
47
48 %% Build a 5-detector several times so see the variation in metrics returned by SGD
49
50 # Solve the model multiple times to see the variations
51 for i in range(5):
52
53     # Train the model and return the predictions made by SGD
54     sgd_clf.fit(X_train, Y_train_5)
55     Y_train_pred = sgd_clf.predict(X_train)
56
57     # Use SK learn model to return metrics
58     accuracy = sk.metrics.accuracy_score(Y_train_5, Y_train_pred)
59     precision = sk.metrics.precision_score(Y_train_5, Y_train_pred)
60     recall = sk.metrics.recall_score(Y_train_5, Y_train_pred)
61     print('accuracy is '+str(np.round(accuracy,4))+'; precision is '+str(np.round(
precision,4))+
62           '; recall is '+str(np.round(recall,4)))
63
64
65 %% Build a 5-detector using k-fold cross-validation

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66
67 # From our example we can see quite a variation in results for a model, making it
68 # hard to select the proper model. However, k-fold cross-validation can help with this.
69
70
71 # make a prediction using the k-fold method to split up the data set. Again,
72 # solve the model multiple times to see the variations
73 for i in range(5):
74
75     # Train the model and return the predictions made by SGD
76     Y_train_pred = sk.model_selection.cross_val_predict(sgd_clf, X_train, Y_train_5, cv=3
77 )
78
79     # Use SK learn model to return metrics
80     accuracy = sk.metrics.accuracy_score(Y_train_5, Y_train_pred)
81     precision = sk.metrics.precision_score(Y_train_5, Y_train_pred)
82     recall = sk.metrics.recall_score(Y_train_5, Y_train_pred)
83     print('accuracy is '+str(np.round(accuracy,4))+'; precision is '+str(np.round(
84 precision,4))+
85         '; recall is '+str(np.round(recall,4)))
```